

67 SCID is an inherited condition in humans.

In one type of SCID, the white blood cells are unable to make the functional enzyme ADA, which is necessary for these cells to divide by mitosis during an immune response.

Scientists have developed a gene therapy treatment whereby a gene is inserted into the DNA of stem cells taken from the bone marrow of a person with this condition. These cells can then be returned to the body of the person, and can divide and differentiate into white blood cells. If this gene therapy is successful, the number of white blood cells should increase significantly.

Which of the following statements correctly describe(s) how this method of gene therapy might work to help a person with this type of SCID?

- 1** Genetically altered stem cells differentiate into white blood cells that are able to produce functional ADA.
 - 2** Genetically altered stem cells can differentiate into gametes so that offspring will not inherit SCID.
 - 3** The sequence of the bases in the DNA of the white blood cells, derived from the genetically altered stem cells, has changed.
- A** 1 only
- B** 2 only
- C** 3 only
- D** 1 and 2 only
- E** 1 and 3 only
- F** 2 and 3 only
- G** 1, 2 and 3